

Learning Objective: To apply concepts such as Asymmetric Information, Adverse Selection, and Moral Hazard to capital market agents, including autonomous AI systems.

Video:

Agentic_AI_&_Agency_Problem

<https://youtu.be/UPieeiwxrFc>

Risks of Agentic AI by IBM Technology

<https://www.youtube.com/watch?v=v07Y4fmSi6Y>

Part 1: Multiple Choice Questions (MCQs)

1. In the context of the Principal–Agent problem, who is typically the "agent" in a corporation?
 - A. The stockholders who own the firm.
 - B. The managers who control the firm's daily activities.
 - C. The government regulators who oversee the market.
 - D. The customers who buy the firm's products.
2. According to lecture slides, the "Principal–Agent Problem" is a specific type of:
 - A. Adverse selection that occurs before a contract is signed.
 - B. Conflict of interest that only happens in the bond market.
 - C. Moral hazard because managers may act in their own interest rather than the owners' interest.
 - D. Free-rider problem involving private information production.
3. The video mentions "underspecification" as a risk, where AI is given a goal but no explicit instructions on how to reach it. In economic terms, this increases the agent's ability to engage in:
 - A. Costly state verification.
 - B. Moral hazard, as the agent may choose paths that benefit itself but harm the principal.
 - C. Screening to reduce asymmetric information.
 - D. Credit rationing to limit risk.

Part 2: True or False

Circle T for True and F for False and provide brief justification.

1. **T / F:** Asymmetric information is a situation where one party has insufficient knowledge about the other party involved in a transaction.
2. **T / F:** The Principal–Agent problem would mitigate if managers (agents) owned 100% of the firm.
3. **T / F:** Agentic AI reduces risk because it always follows human ethical values without the need for technical guardrails.
4. **T / F:** "Costly state verification" refers to the expensive process of monitoring an agent's activities to reduce moral hazard.

Part 3: Matching

Match the Economic Concept from lecture slides (Column A) with the AI Risk/Governance tool from the video (Column B).

Column A: Economic Concept	Column B: AI Context
1. Monitoring	A. Auditability: The ability to trace back how an AI arrived at a decision.
2. Restrictive Covenants	B. Bad Actors: People trying to make an agent act against an organization's policies.
3. Adverse Selection	C. Human in the Loop: Requiring human approval before an AI takes a specific action.
4. Intermediation	D. Technical Guardrails: Rules that prevent an AI from going outside its predefined areas.

Part 4: Ordering

The following are steps in a "Moral Hazard" scenario involving an investment manager or an autonomous AI agent. Put them in the correct order (1-4).

- (___) The agent (manager/AI) takes an excessively risky action because they there are grey areas in what does risk mean.
- (___) The principal (investor) provides funds to the agent.
- (___) The principal suffers a financial loss while the agent remains protected or benefits from the risk.
- (___) Information becomes asymmetric because the principal cannot perfectly monitor the agent's daily choices.